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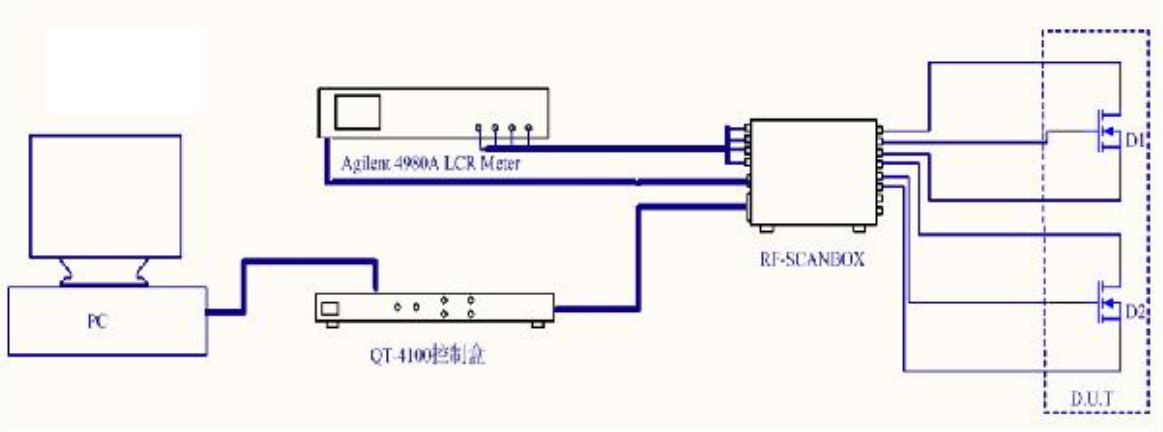
硬件连接示意图
Hardware connection diagram



1.启动时有先后顺序，流程为:LCR模组上电->CB01控制盒上电->打开Loger。
1. The startup sequence is as follows: Power on LCR modules -> Power on the CB01 control box -> Open the Loger.

2.LCR属于精密测量仪器，为保证精度及稳定性必须注意测试环境的干扰影响。
2.LCR is a precision measuring instrument. In order to ensure the accuracy and stability, attention must be paid to the interference of the test environment.

校准操作说明

由于是使用高频交流信号作为施加源，测试端口线及测试片的不同，受分布参数的影响导致测试出来的结果与真实值会有一定的偏差；一些MOS产品，特别输入电容小于1nF的产品，测量时高频信号源的相位容易角度出现微小偏差，这时也会对测量结果会有一定偏差；对低阻抗器件测试时，此误差对测量结果影响较大。当出来这些偏差时，可以执行校准修正程序修正测量结果。

采用 RF_ScanBox +E4980进行测量 RG ，CISS，CRSS，COSS 等项目时，可以执行校准修正程序进行修正测量结果。

当更换测试端口线和测试座的改变时，建议重新OPEN/SHORT校准。

一些MOS产品，特别输入电容小于1nF的产品，建议OPEN/SHORT/LOAD校准。

Because the high frequency AC signal is used as the application source, the test port line and the test piece are different, and the distribution parameters will lead to a certain deviation between the test result and the real value. For some MOS products, especially those whose input capacitance is less than 1nF, the phase Angle of high-frequency signal source is prone to slight deviation during measurement, and the measurement results will be biased to a certain extent. When testing low impedance devices, this error has a great impact on the measurement results. When these deviations occur, a calibration correction procedure can be performed to correct the measurement results.

When using RF_ScanBox +E4980 to measure RG, CISS, CRSS, COSS and other items, the calibration correction procedure can be carried out to correct the measurement results.

When replacing test port lines and test seat changes, it is recommended to re-open /SHORT calibration.

OPEN/SHORT/LOAD calibration is recommended for some MOS products, especially those with input capacitance less than 1nF.

PART 1

单DIE校准操作说明
Single DIE calibration operation instructions

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二. 单DIE校准说明

RF_ScanBox+E4980 校准修正分为“开路修正”，“短路修正”，“标准负载修正”。
开路修正LCRopenCal：主要用于修正测试高阻抗时的偏差，补偿由测试夹具引起的杂散导纳。
短路修正LCRshortCal：主要用于修正测试低阻抗时的偏差，补偿由测试夹具引起的剩余阻抗。
负载修正LCRloadCal：主要用于使用阻抗特性较为复杂的测试夹具测量修正，引入标准值进行修正。

注：
1、 编写测试程序时必须在修正的测试项之前增加 RF_Scanbox 测试项进行设定RFSCANBOX 的输出端,否则将无法校准修正(logger 还是会显示“1”)。
2、 校准修正环境尽可能接近生产测试的环境，减少偏差。
3、 所有端口线的远端屏蔽层必须短在一起，尽可能使端口线等效阻抗是 50 欧。
4、 执行相关的校准修正后，默认测试时叠加校准修正值。如果不需要叠加校准修正值，可以在E4980A 面板上手动取消（参考 agilent E4980A 说明书）

在使用RF_ScanBox中间转接接到生产测试座进行校准操作时，校准程序最开始需要使用RF_ScanBox项目，根据器件对应引脚，用于进入 RF_Scanbox 设定界面进行RF_Scanbox输入与输出切换连接设定（测量点的设定）。

单DIE校准操作说明

二.Single DIE calibration instructions

RF_ScanBox+E4980 Calibration correction includes "open circuit correction", "short circuit correction", "standard load correction".

Open Circuit correction LCRopenCal: It is mainly used to correct the deviation when testing high impedance and to compensate for stray admittance caused by the test fixture.

Short-circuit correction LCRshortCal: Mainly used to correct the deviation when testing low impedance, to compensate for the remaining impedance caused by the test fixture.

Load correction LCRloadCal: It is mainly used for measurement correction of test fixtures with more complex impedance characteristics, and the standard value is introduced for correction.

Note:

1. When writing the test program, you must add the RF_Scanbox test item before the corrected test item to set the output end of the RFSCANBOX; otherwise, the correction cannot be calibrated (logger still displays "1").
2, calibration and correction environment as close as possible to the production test environment, to reduce the deviation.
3. The shielding layer at the far end of all port lines must be short together, so that the equivalent impedance of port lines is 50 ohMs as far as possible.
4. After the relevant calibration correction is performed, the calibration correction is superimposed during the default test. If stacking calibration correction is not required, it can be manually cancelled on the E4980A panel (refer to agilent E4980A instruction manual).

When the RF_ScanBox intermediate switch is connected to the production test stand for calibration operation, the calibration procedure begins with the RF_ScanBox item, which is used to enter the RF_Scanbox setting interface for RF_Scanbox input and output switching connection setting (setting of measuring points) according to the corresponding pin of the device.

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二.1. Open校准LCRopenCal

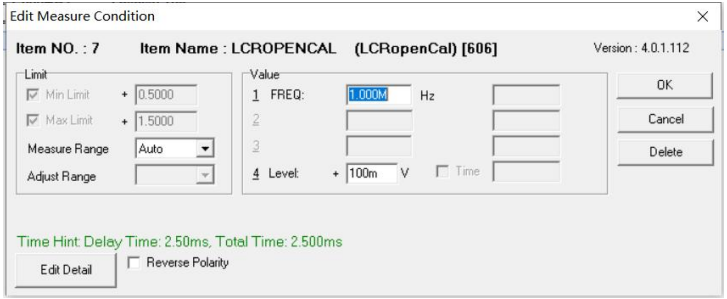
在进行开路修正数据测量必须使测试端处于开路状态（即对应每个单通道的 Forcing 和 Sense 需要短接）。开路修正数据测量过程中不要接触测量夹具或靠近测量夹具，不能压有正常产品。手动测试一次以上，看返回数值，如果为“1”即说明校准修正成功，如果为“0”，即校准修失败。

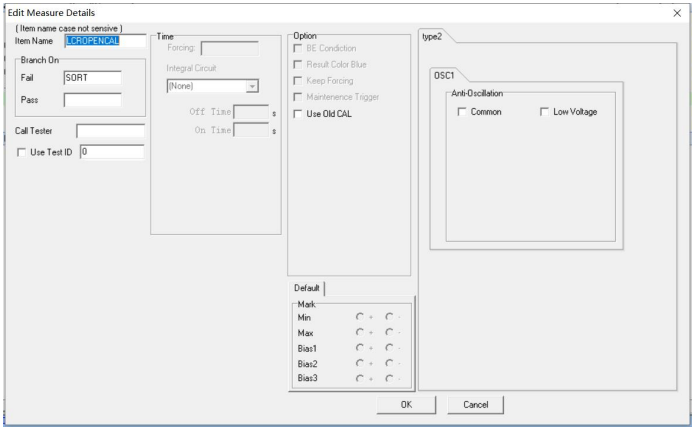
Measure Rang：用于选择测量档位，选择的电阻根据设置Level幅度判断测量的电流档位，因不清楚电容实际大小即流经电流大小，一航默认为AUTO自动选档。

Level：给D.U.T 施加交流偏置的峰值电压。

FREG：给D.U.T 施加的交流偏置频率，默认值为1MHz。

（注意：校准时设置Measure Rang、Level、FREG参数必须要与LCR_RG、CLR_CISS、LCR_COSS、LCR_CRSS测试使用的Measure Rang、Level、FREG一致）





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二.1. Open calibration LCRopenCal

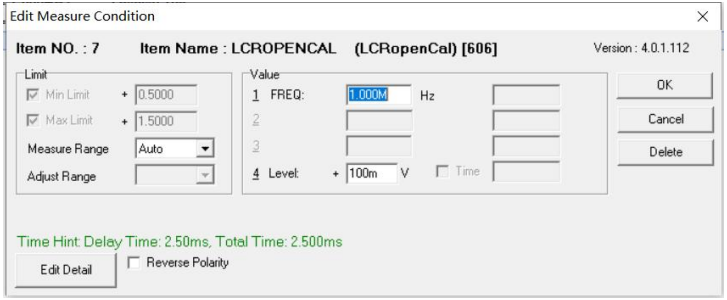
In conducting open-circuit correction data measurement, the test end must be in an open-circuit state (i.e. Forcing and Sense the need for short-circuit corresponding to each single channel). Do not touch or approach the measuring fixture during open circuit correction data measurement. Do not press normal products. After manual test for more than one time, check the returned value. If it is "1", it means successful calibration correction; if it is "0", it means failed calibration correction.

Measure Rang: It is used to select the measuring gear, and the selected resistance judges the measured current gear according to the set Level range. Because it is not clear that the actual size of the capacitor, namely the size of the flowing current, the default is AUTO for the first flight.

Level: Peak voltage to apply AC bias to D.U.T.

FREG: AC bias frequency applied to D.U.T., default is 1MHz.

(Note: The Measure Rang, Level and FREG parameters set during calibration must be consistent with the Measure Rang, Level and FREG used in LCR_RG, CLR_CISS, LCR_COSS and LCR_CRSS tests)



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二.2. SHORT校准LCRshortCal

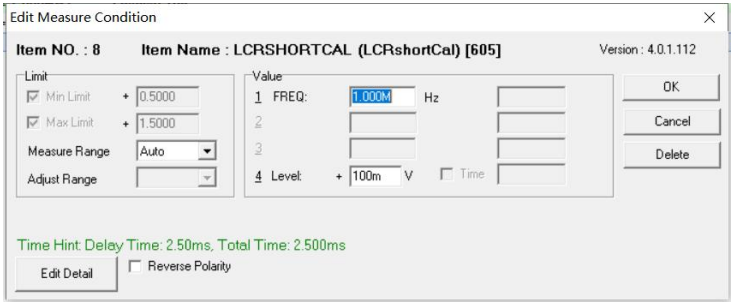
在进行短路修正数据测量必须使用与被测器件相同型状的导电体进行将所用的端口进行短接（即对应全部通道的 Forcing 和 Sense 需要短接一起），不能用正常产品。手动测试一次以上，看返回数值，如果为“1”即说明校准修正成功，如果为“0”，即校准修失败。

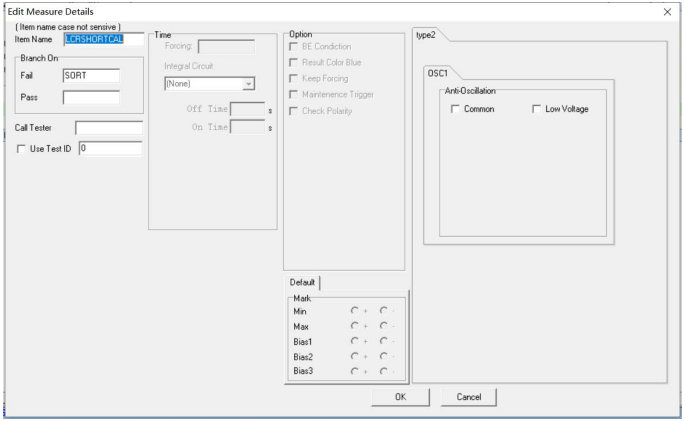
Measure Rang：用于选择测量档位，选择的电阻根据设置Level幅度判断测量的电流档位，因不清楚电容实际大小即流经电流大小，一航默认为AUTO自动选档。

Level：给D.U.T 施加交流偏置的峰值电压。

FREG：给D.U.T 施加的交流偏置频率，默认值为1MHz。

（注意：校准时设置Measure Rang、Level、FREG参数必须要与LCR_RG、CLR_CISS、LCR_COSS、LCR_CRSS测试使用的Measure Rang、Level、FREG一致）





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二.2. SHORT calibration LCRshortCal

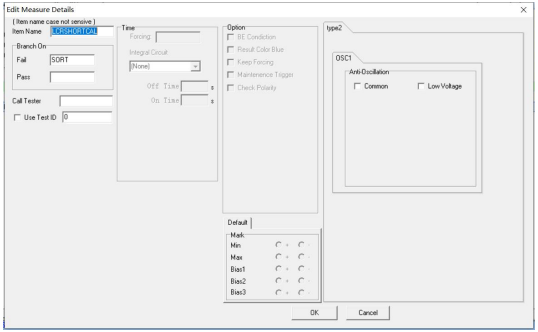
In short-circuit correction data measurement must use the same type of conductive body as the device being measured to short the used ports (that is, Forcing all channels to be forced and Sense needs to be short together), cannot use normal products. After manual test for more than one time, check the returned value. If it is "1", it means successful calibration correction; if it is "0", it means failed calibration correction.

Measure Rang: It is used to select the measuring gear, and the selected resistance judges the measured current gear according to the set Level range. Because it is not clear that the actual size of the capacitor, namely the size of the flowing current, the default is AUTO for the first flight.

Level: Peak voltage to apply AC bias to D.U.T.

FREG: AC bias frequency applied to D.U.T., default is 1MHz.

(Note: The Measure Rang, Level and FREG parameters set during calibration must be consistent with the Measure Rang, Level and FREG used in LCR_RG, CLR_CISS, LCR_COSS and LCR_CRSS tests)



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二.3. Load校准LCRloadCal

在进行标准负载修正数据测量必须使用 将被测器件参数相近的器件（标准样管），与测试端进行连接。手动测试一次以上，看返回数值，如果为“1”即说明校准修正成功，如果为“0”，即校准修失败。

Measure Rang：用于选择测量档位，选择的电阻根据设置Level幅度判断测量的电流档位，因不清楚电容实际大小即流经电流大小，一航默认为AUTO自动选档。

VDS：给D.U.T 施加的直流偏置电。

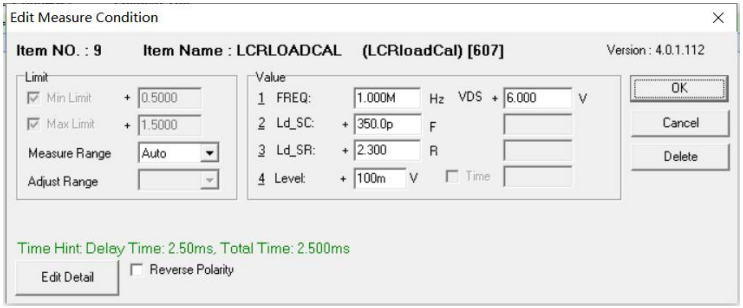
Level：给D.U.T 施加交流偏置的峰值电压。

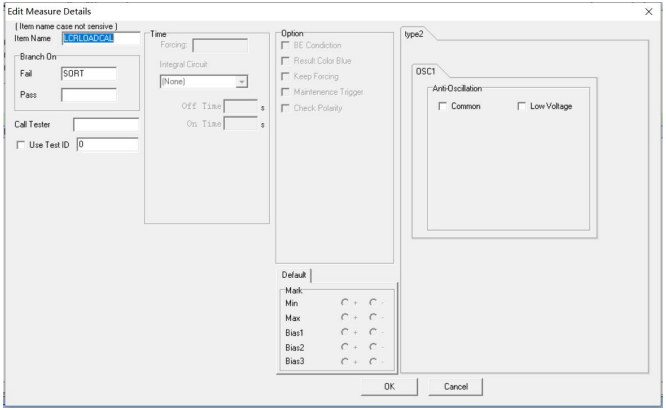
FREG：给D.U.T 施加的交流偏置频率，默认值为1MHz。

Ld_SC： 填入标准负载的等效电容。

Ld_SR： 填入标准负载的等效电阻。

（注意：校准时设置Measure Rang、Level、FREG、VDS参数必须要与LCR_RG、CLR_CISS、LCR_COSS、LCR_CRSS测试使用的Measure Rang、Level、FREG、VDS一致）





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二. 3. Load calibrates LCRloadCal

The device (standard sample tube) with similar parameters of the device under test must be connected to the test terminal when the standard load correction data is measured. After manual test for more than one time, check the returned value. If it is "1", it means successful calibration correction; if it is "0", it means failed calibration correction.

Measure Rang: It is used to select the measuring gear, and the selected resistance judges the measured current gear according to the set Level range. Because it is not clear that the actual size of the capacitor, namely the size of the flowing current, the default is AUTO for the first flight.

VDS: DC bias applied to D.U.T.

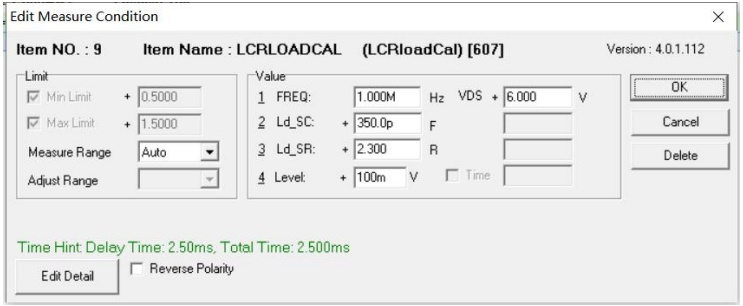
Level: Peak voltage to apply AC bias to D.U.T.

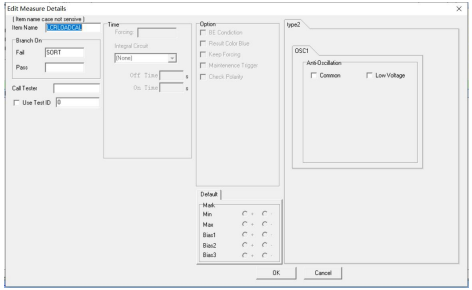
FREG: AC bias frequency applied to D.U.T., default is 1MHz.

Ld_SC: Equivalent capacitance filled in with standard load.

Ld_SR: Equivalent resistance to fill in a standard load.

(Note: The parameters of Measure Rang, Level, FREG and VDS set during calibration must be consistent with the measures Rang, Level, FREG and VDS used in LCR_RG, CLR_CISS, LCR_COSS and LCR_CRSS tests)





PART 2

其他说明Other instructions

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初始化E4980校准数据 E4980AReset

对E4980A初始化校准数据，即清除E4980A保存的校准数据。

注意执行 E4980AReset 需要较长的时间，约需 30S 以上（可以留意 E4980A 的屏幕是否恢复为测量界面）。这复位清除期间请不要做任何的动作。

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Example Initialize E4980 calibration data E4980AReset

Initialize the calibration data of the E4980A, that is, clear the calibration data saved by the E4980A.

It takes 30 seconds to execute the E4980AReset (Check whether the screen of the E4980A is restored to the measurement screen). Please do not do anything during the reset clearance.

其他说明

Monitor信息：

```
24 11:11:19 ************************************************************
24 11:11:19 firmware Build Date:Nov 24 2021   Time:18:02:26
24 11:11:19 This firmware is built for ControlBOX Tester
24 11:11:19 TesterController board Memroy:8192KB
24 11:11:19 Flash info, Device_Manufacture ID(Register):0000_0000_0000_0000
DirectReadFlash:0001_2249_0000_0000
24 11:11:19 FlashID unknown error, set flash size to 2M.
24 11:11:19 FLASH_SPAN:2048 KB
24 11:11:21 Firmware MaxSize:1920k
24 11:11:21 Firmware write to flash Date:2021-11-24 18:02:32 ver:05.1093
24 11:11:21 HardwareInfo(31):SDRG0018039
24 11:11:21 now access ssram only!
                .
                .
24 11:11:26 ************************************************************
24 11:11:26 Station <A> MeasureItem:6 SortItem:5 has been loaded!
24 11:11:26 FileName:C:\Users\PowerTECH\Desktop\20211224验证DEMO机SDRG0020128COSS
\CRTS095N10N-AA-X3-E-FT-RGCG.ptf
24 11:11:26 Station <A> Station: 1. GPIB SBOX Exist. ID:33074
24 11:11:26 RFSCANBOX VDS max_output:40V
24 11:11:26 ScanBoxType [S: 0] [B: 0] [C: 0] [E: 0]
24 11:11:26 Station <A> Fast Measure Data has been loaded! num:8
24 11:11:26 ************************************************************
24 11:11:26 Station <A> loading sort limit options...
```

固件版本
Firmware version

GPIB SBOX ID

Scanbox信息

谢谢

THANK YOU

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